Virtual Leader-Follower Based Platooning Under Mixed Traffic: A Data-Driven Distributionally Robust MPC Method

Fengkun Gao, Bo Yang, *Senior Member, IEEE*, Chao Ning, *Senior Member, IEEE*, and Xinping Guan, *Fellow, IEEE*

***Abstract*—The trajectory prediction information of human-driven vehicles (HDVs) is imperative for the platooning of connected and automated vehicles (CAVs) in mixed traffic flow. However, due to the inherent uncertainty of HDVs, prediction errors are inevitable and time-varying, which results in inefficient control actions and even safety issues. To address the problem, we propose a virtual leader-follower platooning strategy to deal with time-varying prediction uncertainties. The controlled platoon tracks the efficient trajectory planned by the virtual leader to avoid unnecessary control inputs and improve speed smoothness. The challenges mainly lie in the modeling of time-varying prediction uncertainties and the design of control strategy for the virtual leader. First, a second-order moment based ambiguity set is employed to describe the distributional uncertainties of trajectory prediction, with parameters identified and updated by the Gaussian Mixture Model-Hidden Markov Model (GMM-HMM) to adapt to the time-varying environments. Further, a distributionally robust model predictive control (DR-MPC) method for the virtual leader is designed, where conditional value-at-risk constraint is adopted to achieve flexible tracking. By exploiting constraint tightening technology, the efficient solution of the DR-MPC problem is obtained. Simulation results demonstrate that our proposed method can significantly avoid undesired control inputs and improve driving smoothness.**

***Index Terms*—Cooperative adaptive cruise control, data-driven control, distributionally robust model predictive control, mixed traffic flow.**

I. Introduction

WITH the development of vehicle-to-everything (V2X) communication and autonomous driving technologies, the platooning of connected and automated vehicles (CAVs) has recently received considerable attention [1], [2]. Based on information sharing, CAVs in the platoon can maintain a smaller inter-distance to reduce wind resistance and improve energy efficiency [3], [4], [5]. In mixed traffic scenarios, i.e., the coexistence of CAVs and human-driven vehicles (HDVs), the inherent uncertainties of HDV have a great effect on the operation of CAV platoon. Additionally, the time-varying traffic environments aggravate the problem. It is significant to investigate the efficient control strategy of CAV platoon to guarantee traffic safety and comfortableness under uncertainties of mixed traffic.

Considering the physical constraints and diverse performance requirements in platoon control, the model predictive control (MPC) method has been widely utilized. Besides, sliding mode control [6] and control barrier function [7] methods have also been investigated to guarantee driving safety. Most traditional MPC methods assume that the trajectory prediction of the preceding vehicle is accurate [8], [9], [10], [11]. However, it is hard to obtain accurate speed and position trajectories of non-connected HDVs in mixed traffic scenario. Typically, this information can be calculated based on acceleration prediction technologies, such as autoregressive model [12] and deep learning [13], [14]. Due to the spatio-temporal nature of traffic environment and diversity of driving styles, the prediction errors are uncertain and time-varying, which challenges the platoon control [15], [16], [17], [18]. Robust MPC (RMPC) and stochastic MPC (SMPC) are commonly used methods for such an uncertain system [19], [20], [21]. In [22], a tube-based MPC control strategy is proposed to ensure driving safety. In the presence of bounded external disturbance, a sequence of minimum robust positive invariant set is obtained by solving an optimization problem, and the actual trajectory of the CAV platoon is bounded by a tube. A min-max MPC with causal disturbance feedback is proposed in [23], where the worst-case control under uncertainty is considered. However, RMPC only utilizes the boundary information of the uncertainty, e.g., the maximum and minimum values of uncertainties. To ensure that state and input constraints are satisfied in all possible cases, the control decisions are usually too conservative.

By introducing probability distribution information, SMPC allows constraint violations, which is beneficial to improving control performance [21], [24], [25]. In [26], a distributed SMPC method is proposed. When the distributional information of uncertainty is available, low-conservative CAV platoon control decisions can be developed based on constraint tightening

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technology. Considering the massive amount of traffic data, a learning-based SMPC control strategy is proposed in [27], which utilizes Gaussian process to learn the dynamics of HDV and estimates the uncertainty of speed prediction. In [28], neural network is utilized to predict the probability of cut-in and an SMPC controller is designed to handle dangerous cut-in maneuvers. However, the above SMPC methods heavily depend on the assumption that the uncertainty probability distribution is compleblacktely available. In practice, only part of the probability information can be obtained, such as mean and variance of the uncertainty. Hence, the emerging distributionally robust control has been investigated in platoon control problem [29], [30]. Compared to RMPC and SMPC, the partial distribution information of random variables can be incorporated ambiguously by distributionally robust optimization, which shows excellence in processing the ambiguity of uncertainties [31]. However, the ambiguity sets of the exiting works are often obtained through offline calculation, which can not adapt to the variations of traffic environment.

Based on the above observations, we aim to develop an efficient platoon control strategy under time-varying uncertainties of mixed traffic. The challenges mainly lie in three aspects. First, how to use massive traffic data to accurately extract the characteristics of prediction uncertainty. At the same time, this information should be updated online with time-varying environment. Second, how to design a flexible car following strategy to adapt to the speed variations of preceding HDV. Finally, how to guarantee the efficient solution of control optimization problem and meet the real-time requirements.

In this paper, we propose a virtual leader-follower platooning strategy for the scenario of mixed traffic, where a data-driven distributionally robust model predictive control (DR-MPC) method is designed for the virtual leader to achieve flexible platoon control. The main contributions are summarized as follows:

- A Gaussian Mixture Model-Hidden Markov Model (GMM-HMM) is utilized to identify the hidden driving states of HDVs and dynamically update the ambiguity set to incorporate partial distribution information of prediction uncertainties.
- By incorporating a conditional value-at-risk (CVaR) constraint, DR-MPC method is designed for virtual leader control to provide a smooth and efficient trajectory. The actual platoon is controlled by distributed model predictive control (DMPC) method.
- An efficient solution of DR-MPC is obtained by exploiting constraint tightening technology. Simulation results demonstrate that the proposed strategy significantly enhances speed smoothness and driving comfortableness.

The remainder of this paper is organized as follows. Section II describes the focused scenario and the control model. The designed data-driven uncertainty modeling method is given in Section III. The proposed flexible tracking strategy and platoon control method are presented in Section IV. Simulation results are given in Section V. Section VI concludes the whole paper. Acknowledgment is provided in VII.

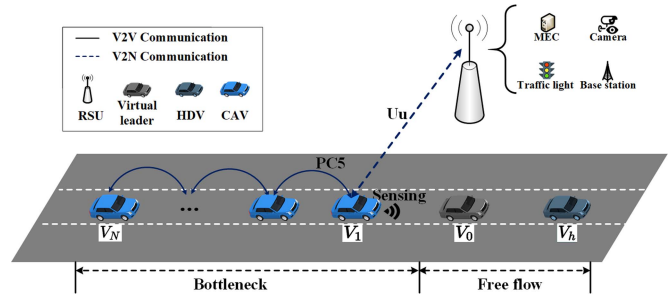

Fig. 1. Platoon control.

II. System Description and Modeling

A. Overview

1) Scenario Description: In this paper, we consider the longitudinal control problem of CAV platoon in the mixed traffic scenario as shown in Fig. 1, where a platoon consisting of N CAVs $V_0, \ldots, V_{N-1}$ follows a non-connected HDV V_h. This means that CAV platoon can not acquire dynamic information of HDV by establishing communication links. Each CAV in the platoon is assumed to be equipped with an on-board unit (OBU), which enables sensing, computing and communication. A virtual leader V_v is added artificially to mitigate the negative effect of HDV uncertainties on the platoon. Based on historical sensing data, the virtual leader can predict the acceleration of the preceding HDV. Further, the future speed and position trajectory of HDV can be calculated by kinematic equation. The control inputs of virtual leader can be developed by incorporating the real-time operation states and prediction information. Considering the dynamic car-following environment, such as variations of traffic density and switching HDV leader, the acceleration prediction error is uncertain and time-varying.

CAV platoon can be formed based on different information flow topologies, such as predecessor-following (PF) and predecessor-leader-following (PLF) [32]. In this paper, the PF topology is adopted, because it can meet the performance requirements with low communication cost. As shown in Fig. 1, each following CAV solely needs to consider the information of its predecessor and maintains predefined inter-vehicle distance. In addition, the leading CAV V_h can also establish connections with the roadside union (RSU) through vehicle-to-network (V2N) communication over the Uu interface to obtain traffic information.

2) Architecture of Virtual Leader-Follower Platooning: The architecture of the proposed virtual leader-follower platooning is shown in Fig. 2. The virtual leader V_v of is responsible for predicting the acceleration of preceding HDV. The prediction results are utilized to construct ambiguity set and develop control inputs u_v based on DR-MPC controller. The real leader of the platoon will track the trajectory of virtual leader instead of the HDV V_h. Hence, the speed smoothness of the whole platoon can be improved. With the aid of V2V communication, each following CAV receives the dynamic information of its predecessor and maintains a fixed time headway based on DMPC controller.

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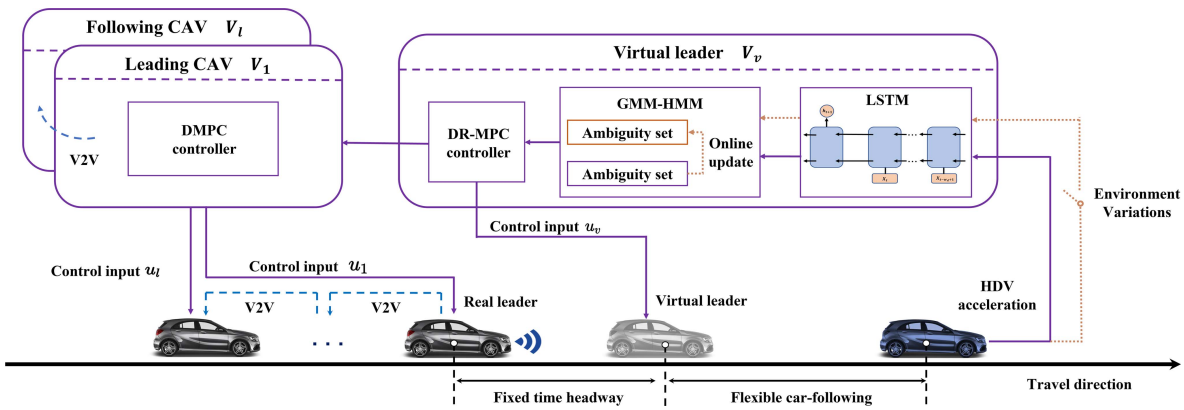

Fig. 2. Architecture of the proposed virtual leader-follower platooning.

When traffic environment varies, such as the variations of traffic density, the ambiguity set will be updated based on newly collected data. Therefore, the DR-MPC controller can adapt to time-varying uncertainties.

B. Control Model

In this section, a discrete linear state space model is given to describe the dynamics of CAV. Assuming that the sampling interval is τ and the current time slot is t, the model is described as follows:

$$s_i(t+\tau) = s_i(t) + \tau v_i(t) + 0.5\tau^2 u_i(t) \tag{1}$$

$$v_i(t+\tau) = v_i(t) + \tau a_i(t),\ i = 0, \ldots, N-1, \tag{2}$$

where s_i, v_i and a_i represent the position, velocity and acceleration of CAV i, respectively. τ is the sampling interval. We define acceleration a as the control input u to regulate the movement of CAV. The control input u is subject to the following physical conditions

$$a_{\min} \leq u(t) \leq a_{\max}. \tag{3}$$

The acceleration of the preceding HDV can be predicted by various methods, such as autoregressive model and deep learning. In this paper, a classical deep learning model, i.e., long-short term memory (LSTM), is adopted. Based on the prediction results, the position and speed can be calculated by the following kinematic equations

$$s_h(t+\tau) = s_h(t) + \tau v_h(t) + 0.5\tau^2(\hat{a}_h(t) + w) \tag{4}$$

$$v_h(t+\tau) = v_h(t) + \tau(\hat{a}_h(t) + w), \tag{5}$$

where s_h and v_h denote the position and speed of HDV V_h, respectively. $\hat{a}_h(t)$ is the predicted acceleration. Stochastic variable w is introduced to describe the uncertainty of acceleration prediction. However, the probability distribution of w is hard to obtain, which poses a great challenge to the design of control strategies for CAV platoon.

For longitudinal control of the platoon, the relative position and speed between the leading CAV and the HDV should be concentrated on, which is defined as follows

$$\Delta s(t) = s_h(t) - s_0(t) \tag{6}$$

$$\Delta v(t) = v_h(t) - v_0(t), \tag{7}$$

Further, the state space model of the car following system can be described as:

$$\Delta s(t+\tau) = \Delta s(t) + \Delta v(t)\tau - 0.5a(t)\tau^2 \tag{8}$$

$$+ 0.5(\hat{a}_h(t) + w(t))\tau^2 \tag{9}$$

$$\Delta v(t+\tau) = \Delta v(t) - a(t)\tau + (\hat{a}_h(t) + w(t))\tau, \tag{10}$$

For simplicity, let $x(t) = [\Delta s(t)\ \Delta v(t)]$, $u(t) = a(t)$, $\hat{\theta}(t) = \hat{a}_h(t)$, the above model can be rewritten in vector form:

$$x(t+\tau) = x(t) + Bu(t) + D(w(t) + \hat{\theta}(t)) \tag{11}$$

The corresponding matrices and vectors are defined as follows

$$A = \begin{bmatrix} 1 & \tau \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} -0.5\tau^2 \\ -\tau \end{bmatrix}, \quad D = \begin{bmatrix} 0.5\tau^2 \\ \tau \end{bmatrix} \tag{12}$$

III. Data-Driven Uncertainty Modeling

A. Second-Order Moment Based Ambiguity Set

Due to the heterogeneity of mixed traffic, statistical characteristics of uncertainties is diverse. Hence, the ambiguity set P under hidden state s is defined as Minkowski sum of several basic ambiguity sets. To incorporate uncertainty information into the design of the control strategy, mean and variance of prediction uncertainties are utilized. The ambiguity set under hidden state s is given by [33]:

$$\mathbb{P}(s) = \sum_{j=1}^{m(s)} \gamma_j(s) \mathrm{P}_j(s)\left(\mathrm{W}, \mu_j(s), \Sigma_j(s)\right) \tag{13}$$

$$\mathrm{P}_j(s)\left(\mathrm{W}, \mu_j(s), \Sigma_j(s)\right) = \left\{ \rho \in \mathrm{M}_+ \;\middle|\; \begin{array}{l} \int_{\mathrm{W}} \rho(d\xi) = 1 \\ \int_{\mathrm{W}} \xi \cdot \rho(d\xi) = \mu_j(s) \\ \int_{\mathrm{W}} \xi\xi^T \cdot \rho(d\xi) \leq \Sigma_j(s) + \mu_j(s)\left(\mu_j(s)\right)^T \end{array} \right\}, \tag{14}$$

where $\mathrm{W} = \{w \mid \mathbf{E}w \leq \mathbf{f}\}$ is the support set of the prediction error. This set describes the boundary information of uncertainty. $m(s)$ is the number of mixture components, and $\gamma_j(s)$ represents

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the occurrence probability of component j. $\mathbb{P}(t)$ represents the ambiguity set at time t. $\mu_j(s)$ and $\Sigma_j(s)$ represent the mean and covariance of the jth mixture component. M_+ is the set of positive Borel measures on $\mathbf{R}^n$, and ρ represents the positive measure. The first constraint in (14) indicates that the probability of the uncertaint on the entire space is 1. The second and third constraints characterize the mean and variance of uncertainty, respectively.

B. Parameters Identification Based on GMM-HMM

In complex traffic environments, the speed variations of HDVs exhibit significant temporal dependence, requiring time series analysis methods to model the prediction uncertainties. Besides, the statistical characteristics of uncertainties can be diverse with time-varying hidden driving styles of drivers. Therefore, this paper employs the GMM-HMM method to predict hidden driving state and identify the parameters of the above ambiguity set 14, i.e., the value of mean $\mu_j(s)$, variance $\Sigma_j(s)$, and weights $\gamma_j(s)$.

The Gaussian mixture model (GMM) is a widely used probabilistic model for data clustering and density estimation [34], [35]. The actual distribution of observed data is described through a linear combination of multiple Gaussian distributions. Specifically, the probability density function of a GMM is a weighted sum of K Gaussian probability density functions:

$$P(y) = \sum_{k=1}^{K} \gamma_k \mathcal{N}(y|\mu_k, \Sigma_k), \tag{15}$$

where y is a D-dimensional random vector, γ_k is the weight coefficient of the k-th Gaussian component. μ_k and Σ_k are the mean and covariance matrix of the k-th Gaussian component, respectively. Each component in the GMM follows the multivariate Gaussian distribution:

$$f\left(y \mid \mu_k, \Sigma_k\right) = \frac{1}{(2\pi)^{\frac{D}{2}} \left|\Sigma_k\right|^{\frac{1}{2}}} e^{-\frac{1}{2}(y-\mu_k)^\top \Sigma_k^{-1}(y-\mu_k)}. \tag{16}$$

The parameters that need to be identified in a GMM include the mean μ_k, covariance Σ_k, and mixture weight coefficients γ_k. These parameters are typically estimated using the Expectation Maximization (EM) algorithm [36].

A Hidden Markov Model (HMM) can be described by a set of hidden states S, a set of observations O, an initial state probability distribution π, a state transition probability matrix A, and an observation probability matrix B [37], [38], which can be denoted by $\lambda = (\pi, A, B)$. The state transition matrix $A = [a_{ij}]$ describes the transition probabilities between hidden states, where $a_{ij} = P(s_{t+1} = j|s_t = i)$ is the probability of transitioning from hidden state i at time t to state j at time $t+1$. The observation probability matrix $B = [b_j(k)]$ describes the probability of observing k given the hidden state j, where$b_j(k) = P(o_t = k|s_t = j)$.

The architecture of the GMM-HMM model is illustrated in Fig. 3. This model is a statistical model that combines GMM and HMM. For a GMM-HMM, the observation distribution for

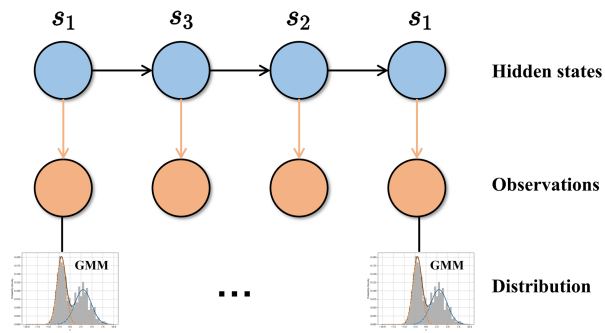

Fig. 3. Architecture of GMM-HMM.

each hidden state is modeled as a GMM, that is

$$b_j(o_t) = \sum_{m=1}^{M} \gamma_{jm} \mathcal{N}(O_t|\mu_{jm}, \Sigma_{jm}), \tag{17}$$

where γ_{jm} represents the weight of the m-th Gaussian component corresponding to the j-th hidden state. μ_{jm} and Σ_{jm} denote the mean and covariance matrix, respectively.

Different human drivers exhibit various driving styles, and the same driver may exhibit different driving behaviors in a complex traffic environment. This paper treats driving behaviors as hidden states within the GMM-HMM, and the acceleration prediction errors are taken as the observation sequence. Based on the historical observations, the parameters $\mu_m(s)$, $\Sigma_m(s)$, and $\gamma_m(s)$ in the ambiguity set can be identified utilizing the GMM-HMM model, which yields the ambiguity set $\mathbb{P}(s)$ under different hidden driving states. Based on the transition between hidden states, the parameters of ambiguity set are updated online, which improves the adaptability of control strategy.

IV. Virtual Leader-Follower Platoon Control

A. Flexible Car-Following Strategy

Due to the stochastic nature of HDVs, if a platoon strictly follows a predetermined speed and position trajectory, it could significantly reduce driving comfortabless and fuel efficiency, and potentially lead to safety issues. Instead, the relative position and relative speed between the virtual leader and the preceding HDV only need to be maintained within a certain range. That is:

$$\mathbf{H}x(t) \leq \mathbf{h}, \tag{18}$$

where $\mathbf{H}$ and $\mathbf{h}$ are given by:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 1 \\ 0 & -1 \end{bmatrix}, \quad \mathbf{h} = \begin{bmatrix} e_s^{\max} \\ -e_s^{\min} \\ e_v^{\max} \\ -e_v^{\min} \end{bmatrix}. \tag{19}$$

However, the above deterministic constraints are often difficult to satisfy. Hence, we employ the distributionally robust conditional value-at-risk (CVaR) constraints on relative speed and position to construct flexible tracking strategy, which is given below:

$$\sup_{\mathbb{P}\in\mathbb{P}(t)} \text{P-CVaR}_{\epsilon_i}\left(\mathbf{H}_i x - \mathbf{h}_i\right) \leq 0, \tag{20}$$

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where ϵ_i is the risk level for the i-th state constraint. This constraint implies that the constraints are satisfied with a probability of at least ϵ_i, which hedges against distributional ambiguity of prediction and penalizes severe constraint violations that could be disruptive to platooning. $\mathbf{H}_i$ and $\mathbf{h}_i$ represent the i-th row of the matrix H and the i-th element of the vector h, respectively. The constraint (20) not only models the uncertainty in the distribution of the prediction error, but also penalizes constraint violations. It forces the relative speed Δv within a predefined set.

When the relative speed satisfies the constraint, the leading CAV has no need to adjust control inputs. Hence, the speed fluctuation of the HDV can be filtered, and the speed smoothness of the platoon can be guaranteed. However, this constraint hinders the efficient solution of the MPC problem. In the following sections, a constraint tightening technology is adopted to reformulate (20).

B. Constraint Tightening

To guarantee the efficient solving process, we reformulate (20) based on the constraint tightening technology proposed in [33]. First, the system state is divided into the nominal state $\bar{x}$ and the prediction error e, that is, $x(k|t) = \bar{x}(k|t) + e(k|t)$. If and only if the nominal system states satisfy the following constraints, the original system states satisfy the distributed robust CVaR constraint.

$$\bar{x}(l|k) \in \bar{X}_k = \{\bar{x}|H\bar{x} \leq h - \eta(k)\} \ominus \left(\oplus_{i=1}^{l-1} A^i DW\right), \quad (21)$$

where $\bar{X}_k$ represents the tightening constraint at time k. $\ominus$ and $\oplus$ represent Pontryagin subtraction and Minkowski addition, respectively. The value of $\eta(t)$ is the solution of semi-definite programming.

$$\min \quad \eta$$

$$\text{s.t. } \varepsilon_i \cdot \beta_i + \sum_{j=1}^{m} \gamma_j \left\{t_{ij} + \mu_j^T \omega_{ij} + \left(\Sigma_j + \mu_j \mu_j^T\right) \Omega_{ij}\right\} \leq 0 \quad (22)$$

$$\begin{bmatrix} \Omega_{ij} & \frac{1}{2}\left(\omega_{ij} + E^T \varphi_{ij}\right) \\ \frac{1}{2}\left(\omega_{ij} + E^T \varphi_{ij}\right)^T & t_{ij} - f^T \varphi_{ij} \end{bmatrix} \geq 0, \forall j \quad (23)$$

$$\begin{bmatrix} \Omega_{ij} & \frac{1}{2}\left(\omega_{ij} - H_i^T + E^T \phi_{ij}\right) \\ \frac{1}{2}\left(\omega_{ij} - H_i^T + E^T \phi_{ij}\right)^T & t_{ij} + \beta_i + \eta - f^T \phi_{ij} \end{bmatrix} \geq 0, \forall j \quad (24)$$

$$\Omega_{ij} \geq 0, \varphi_{ij} \geq 0, \phi_{ij} \geq 0, \forall j \quad (25)$$

where t_{ij}, ω_{ij} and Ω_{ij} are dual variables with respect to the constraints in the ambiguity set. The constraints (23) and (24) are linear Matrix inequality constraints. $\psi_{ij} \geq$ and $\phi_{ij} \geq$ are dual variables with respect to the constraint $\mathbf{E}w \leq \mathbf{f}$.

C. DR-MPC Method

For the virtual leader, the travel safety and speed smoothness of the whole platoon are the main objectives. The time headway strategy is adopted, i.e., $D(t) = hv(t) + d$, where d denotes time is the fixed inter-vehicle distance. Therefore, the position tracking penalty term is defined as:

$$J_1(k) = a_1 \|s_h(k|t) - s_0(k|t) - D(t)\|_2^2 \quad (26)$$

$s_h(k|t)$ and $s_0(k|t)$ represent the position prediction of preceding HDV and virtual leader at time t for time $t + k$, respectively. The speed tracking penalty term is defined as:

$$J_2(k) = a_2 \|v_h(k|t) - v_0(k|t)\|_2^2 \quad (27)$$

Considering the operation of following platoon, the optimal trajectory obtained at the last time step should be matained to avoid frequent variations of the reference trajectory.

$$J_3(k) = \|F \cdot E(x_0(k|t) - x_0^a(k|t))\|_2^2 \quad (28)$$

where $F = [a_{3,1},\ 0;\ 0,\ a_{3,2}]$ is the scaling matrix, and E is identity matrix. The trajectory $x_0^a(k|t)$ is defined by

$$x_0^a(k|t) = \begin{cases} x_0^*(k+1|t-1), & k = 0, 1, \ldots, N_p - 1, \\ A \cdot x_0^*\left(N_p|t-1\right), & k = N_p, \end{cases} \quad (29)$$

To improve the speed smoothness and energy efficiency, the magnitude of acceleration/decelerationed are penalized by

$$J_4(k) = a_4 \|u(k|t)\|_2^2. \quad (30)$$

Therefore, the control sequence of the virtual leader at time t can be obtained by solving the following problem, and the first element of the control sequence is applied to actual control.

$$\mathbf{P1}: \min_{u_l} \sum_{k=0}^{N_p-1} \sum_{i=1}^{4} J_i\left(x_l(k|t), u_l(k|t), x_{l-1}(k|t)\right)$$

$$\text{s.t. } \bar{x}(t+\tau) = A\bar{x}(t) + Bu(t), \quad (31)$$

$$\bar{x}(0|t) = x(t), u_l(k|t) \in \mathcal{U}, \quad (32)$$

$$\bar{x}(k|t) \in \bar{X}_k, \bar{x}(N|t) \in X_f, \quad (33)$$

$$k = 0, \ldots, N-1, \quad (34)$$

where X_f is the terminal set defined by robust positively invariant set of the system. According to the results in [33], the iterative feasibility and asymptotic stability of controller **P1** can be guaranteed. For the real leader and the following vehicles of the platoon, they only need to calculate the control input through standard DMPC, which is based on the future trajectory of their predecessors acquired through V2V communication.

V. Simulation

In the simulations, we consider a platoon consisting of three CAVs. The real leader of the platoon tracks the trajectory of the virtual leader. The sampling interval τ and the prediction horizon N_p are set to 0.1 s and 5, respectively. The constant time headway is selected as $h = 0.5$ s to guarantee travel safety.

A. Trajectory Prediction Results

The real taxi trajectory data, collected from the Inner Ring Road in Shanghai, China, is utility for designing simulations. The dataset contains the traffic data of 65836 road segments

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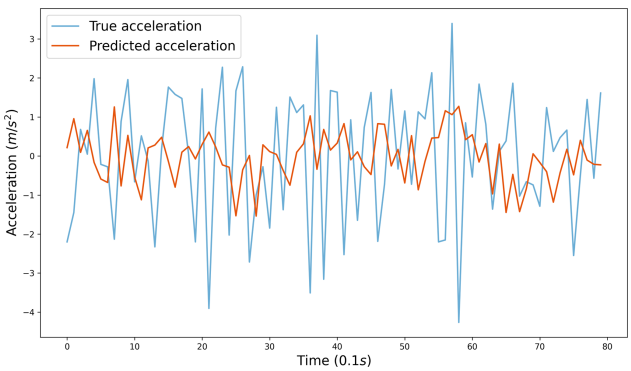

Fig. 4. Acceleration predictive results.

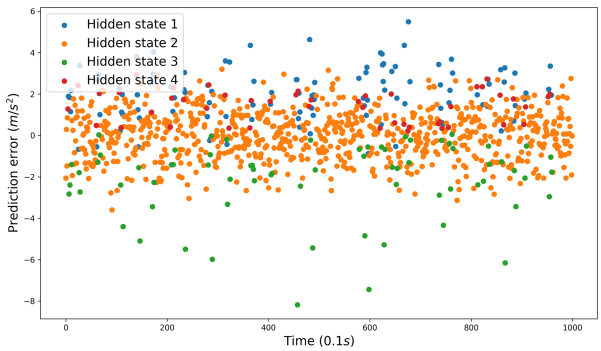

Fig. 5. Comparison of speed trajectories under different control methods.

during April 1-30, 2015, whose size is about 300 G. Vehicle ID, GPS data, velocity, and accidents are recorded.

The classical LSTM algorithm is used to predict the acceleration of the preceding HDV. Subsequently, the speed and position trajectories of preceding HDV can be calculated based on kinematic equations. The length of input time series is 30, and the prediction step length is 5. The LSTM network is constructed using the Sequential model in TensorFlow [39], which includes 4 bidirectional LSTM layers and incorporates Dropout layers to prevent overfitting.

The acceleration prediction results are shown in Fig. 4, where the black line is the true value of acceleration, and the red line denotes the prediction value of the acceleration based on historical data. The time series prediction method, i.e., LSTM, can capture the variation trend of the HDV speed. However, due to the uncertainty of HDV, prediction errors are inevitable, and the errors themselves are uncertain.

B. Prediction Uncertainty Modeling Results

Based on the aforementioned trajectory prediction results, a dataset of acceleration prediction errors can be obtained. The GMM-HMM model is then utilized to extract the stochastic characteristics of human-driven vehicles. To capture the temporal dependencies, the original data is divided into time series data with length of 10 to serve as model input.

The results of prediction errors and hidden driving states over time are shown in Fig. 5. The points represent acceleration prediction errors, and the color of the points indicates different hidden driving states. It is obvious that the hidden driving states

TABLE I
PARAMETERS OF GMM-HMM

Components	Parameter	State 1	State 2	State 3	State 4
Component 1	Mean	2.00	−0.05	−5.48	1.82
	Variance	1.43	1.18	1.85	0.21
	Weight	0.70	0.94	0.11	0.58
Component 2	Mean	0.31	0.70	−1.28	0.49
	Variance	0.87	1.52	1.05	0.02
	Weight	0.30	0.06	0.89	0.42

varying with dynamic traffic environments, which brings uncertainties and challenges to the control method design. According to Fig. 5, the observed data points of different hidden driving states are not completely separated. They may exhibit similar observed values (prediction errors) under different hidden states. However, the distribution characteristics of each class of hidden driving states are distinctly different. The black data points represente hidden state 1, which are mostly distributed in the positive plane. The hidden state 3 are illustrated in green color, which are mostly distributed in the negative plane. The orange data points are evenly distributed on both sides of the zero point.

The main challenge in applying the GMM-HMM model is selecting the number of hidden states and the number of mixture components. To address the problem, we performe multiple experiments to cross-validate the parameters. According to simulation results, the GMM-HMM model is configured with 4 hidden states and 2 mixture components. The fitting results of the observation probability distributions for each hidden state are shown in Fig. 7. The histograms represent the distribution density of the actual data, and the distribution curves of GMM under each hidden state are illustrated in different colors. Although the observed data points of different hidden driving states are not completely separated in Fig. 5, the distribution characteristics of each class of hidden driving states are distinctly different. As shown in Fig. 7, the data distributions of hidden state 1 and hidden state 2 are similar to the normal distribution, which can be considered to be of the "Normal" type. Similarly, hidden state 3 and hidden state 4 can be taken as "Conservative" type and "Aggressive" type, respectively.

The data samples and distribution fitting results for each hidden state are shown in Fig. 6, and the detailed parameters of the GMM-HMM model are summarized in Table I. Fig. 6(a) and (b) illustrate the data samples and distribution fitting results for hidden states 1 and 2, respectively. The data distributions of these two states approximate a normal distribution and can be considered as "Normal" driving states. Although these two driving states belong to the same category, the mean of hidden state 1 is larger than that of hidden state 2, which can be cosidered as distribution shift affected by traffic environments. Hidden state 3 has a negative mean and larger variance, indicating a more dispersed data distribution, which can be considered as a "Conservative" driving state. In contrast, hidden state 4 has a positive mean and smaller variance, indicating a more concentrated data distribution, which can be considered as an "Aggressive" driving state. In practical platoon control, the driving states of the HDV can be identified based on real-time observation data. The parameters of the ambiguity set are updated at the same time to

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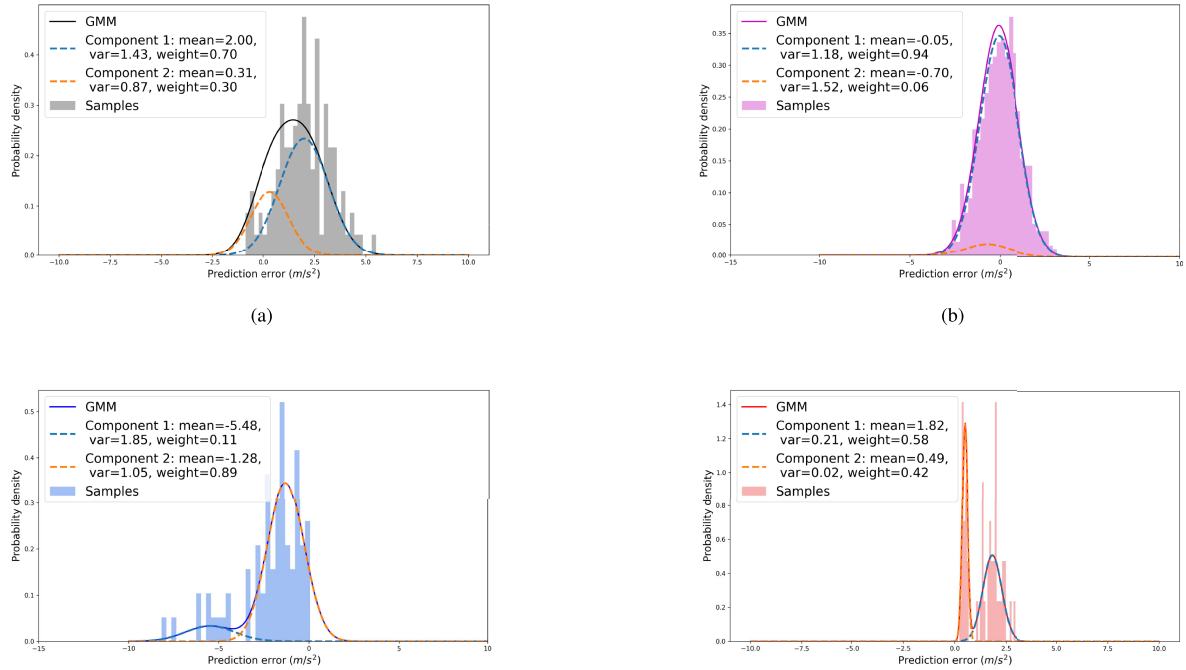

Fig. 6. Data samples and distribution fitting results for each hidden state. (a) Fitting results of hidden driving state 1. (b) Fitting results of hidden driving state 2. (c) Fitting results of hidden driving state 3. (d) Fitting results of hidden driving state 4.

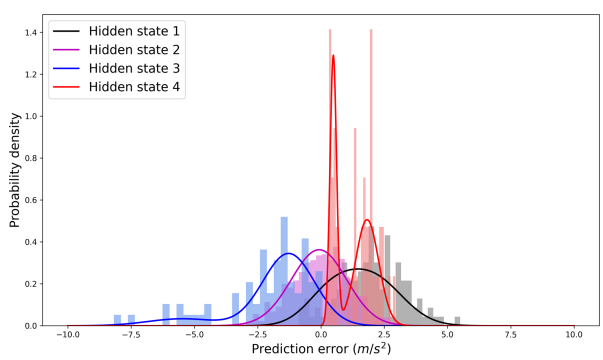

Fig. 7. Observation probability distributions of GMM-HMM.

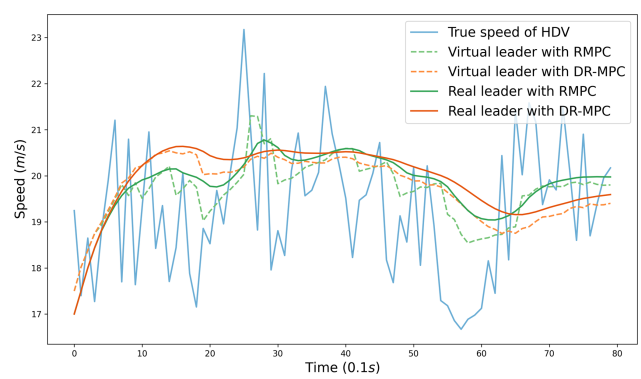

Fig. 8. Comparison of speed trajectories under different control methods.

TABLE II
PARAMETER SETTINGS OF PLATOON CONTROL

Parameter	Symbol	Value	Unit
Time headway of virtual leading vehicle	h_0	1.5	s
Time headway of real platoon	h_l	0.5	s
Control horizon	N_p	5	/
Sampling interval	τ	0.1	s
Lower bound of fixed inter-distance	$d_{\min}$	3	m
Upper bound of fixed inter-distance	$d_{\max}$	5	m

develop platoon control actions, which ensures the adaptability of control strategies in dynamic traffic environments.

C. Control Performance Analysis

The parameter settings are given in Table II. The comparison of speed trajectories under different control methods is shown in the Fig. 8. The black line is the true speed trajectory of the preceding HDV. The dotted lines and solid lines denote the speed trajectories of virtual leader and real leader, respectively. It is obvious that the two control methods can both achieve car-following driving under mixed traffic. Compared with the RMPC method, the proposed DR-MPC method provides smoother speed reference for the platoon. From the speed variations of real leader, unnecessary acceleration and deceleration is avoided and the speed travel comfortableness is improved. Because the DR-MPC introduces distribution information of predition uncertainty, the control decisions has less conservation. Besides, the incremental DPMM model can adapt to the variations of uncertainty. By introducing a virtual leader, the impact of HDV uncertainties on the operation of platoon is mitigated.

The position trajectories of all CAVs within the platoon and preceding HDV are shown in the Fig. 9. The blue line denotes the position trajectory of preceding HDV. Green and red lines represent the position trajectory of leading CAV under DRO-MPC and RMPC, respectively. It is obvious that both control

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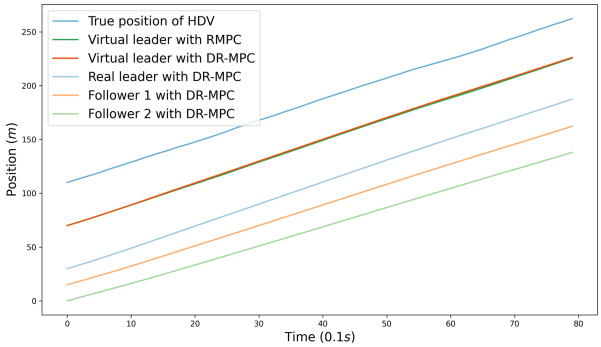

Fig. 9. Comparison of position trajectories under different control methods.

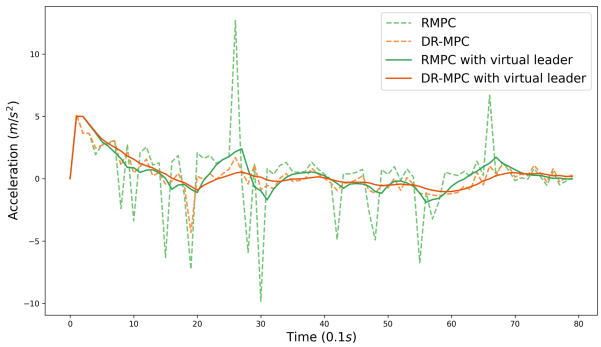

Fig. 10. Control inputs of real leader under different control methods.

methods can maintain the predefined car-following distance and ensure driving safety.

The control inputs of real leader under different control methods is given in Fig. 10. For both proposed virtual leader-follower strategy and conventional strategy, the fluctuations of control inputs under RMPC is more frequent and larger the that of DR-MPC. Because RMPC only utilizes the the boundary information of prediction uncertainty, which leads to overly conservative decisions. Compared with RMPC, DR-MPC introduces distribution information of prediction uncertainty through mean and variance, which enables the virtual leader develop less conservative decisions to improve the smoothness and comfortableness. Besides, the introduction of virtual leader significantly reduce unnecessary control inputs and improve energy efficiency and travel comfortableness.

VI. Conclusion

In this paper, we propose an virtual leader-follower strategy for CAV platoon control under time-varying uncertainties of mixed traffic scenario. Virtual leader provides efficient and smooth reference trajectory for the actual platoon control. First, the uncertainty of HDV are modeled by a second-order moment-based ambiguity set. A GMM-HMM based method is proposed for online updating the parameters of the ambiguity set by identifying hidden driving states. Furthermore, a flexible tracking policy is designed for virtual leader by incorporating conditional value-at-risk constraint, and a DR-MPC method is designed for virtual leader control. An efficient solution is guaranteed with the aid of constraint tightening technology. Finally, simulation results demonstrate that our proposed method can significantly avoid undesired acceleration (deceleration) and improve driving smoothness in time-varying traffic environment.

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